

Industrial Internship Report on "Crop Yield Prediction Using Machine Learning"

**Prepared by
[SUBHANI SHAIK]**

Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was "Crop Yield Prediction Using Machine Learning". The objective of this project was to develop a machine learning model that predicts agricultural crop yield based on factors such as crop type, state, cultivation cost, and production cost. The project involved data collection, data preprocessing, exploratory data analysis (EDA), feature engineering, label encoding, model training using the Random Forest Regressor algorithm, and performance evaluation. The trained model achieved an R^2 score of 0.9463, demonstrating high prediction accuracy. Finally, the model was saved and used to predict crop yield for new user inputs, making it a practical decision-support tool for agricultural planning.

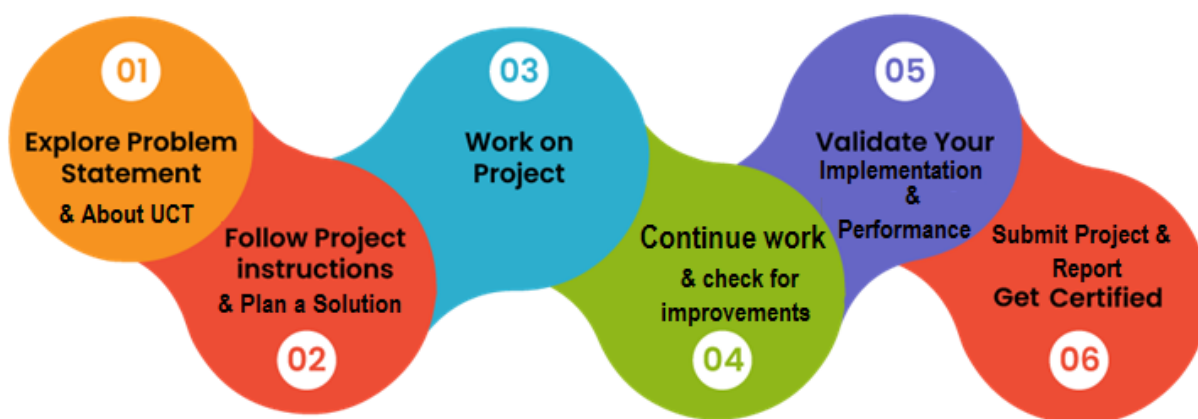
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1 Preface

The Industrial Internship conducted by **upskill Campus** in collaboration with **The IoT Academy** and **UniConverge Technologies Pvt. Ltd. (UCT)** provided an excellent opportunity to gain practical knowledge and hands-on experience in Machine Learning and Data Science. During this six-week internship, I learned how industrial projects are planned, developed, tested, and documented following real-world software development practices.

The internship played an important role in my career development by enhancing my technical skills in **Python, Data Analysis, Machine Learning, Data Visualization, and Model Development**. It helped me understand how theoretical concepts learned in academics are applied to solve practical problems in the agricultural domain.



The project assigned during the internship was "**Crop Yield Prediction Using Machine Learning**." The objective of this project was to develop a machine learning model capable of predicting agricultural crop yield using parameters such as crop type, state, cultivation cost, and production cost. The project involved various stages including data collection, preprocessing, exploratory data analysis, feature engineering, model training using the **Random Forest Regressor** algorithm, performance evaluation, and prediction of crop yield.

The internship program was well organized by UCT and upskill Campus. During the initial phase, we understood the project requirements and problem statement. We then collected and analyzed the dataset, performed data preprocessing, built and evaluated the machine learning model, improved the prediction performance, and finally completed the project documentation and report submission.

Finally, I would like to encourage my juniors and peers to actively participate in internships and real-world projects, as they provide valuable practical experience, improve technical skills, and prepare students for future careers in software development, data science, and machine learning.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



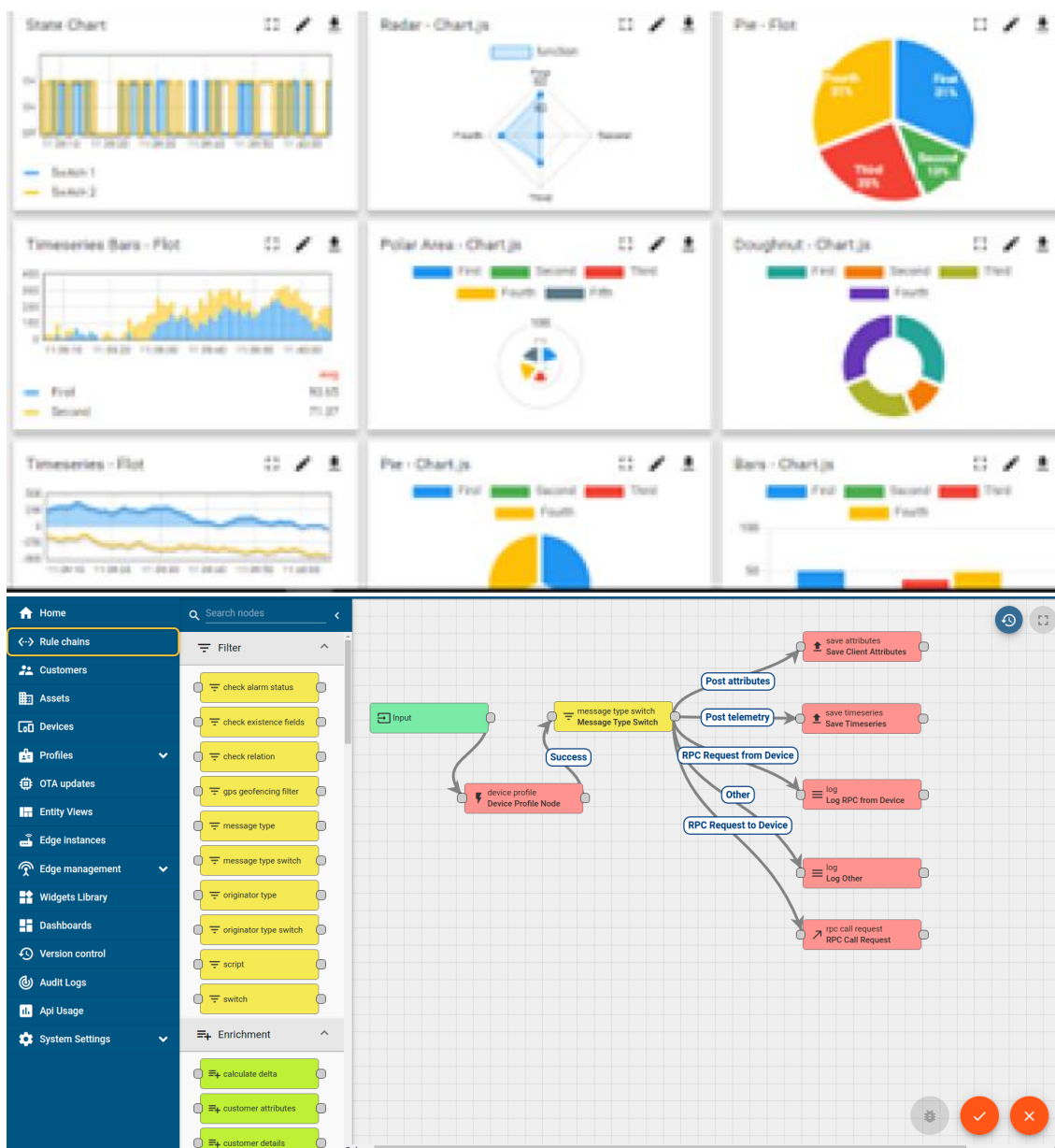
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



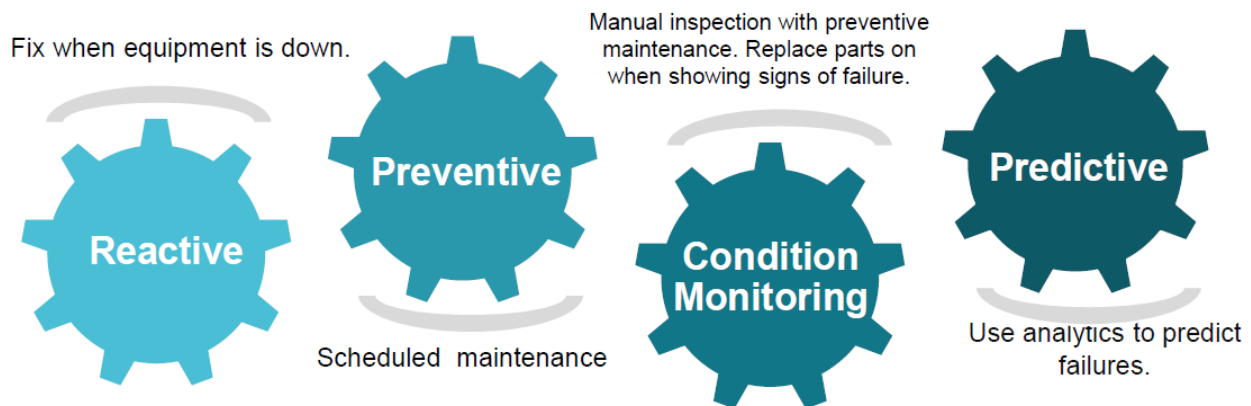


iii. based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.

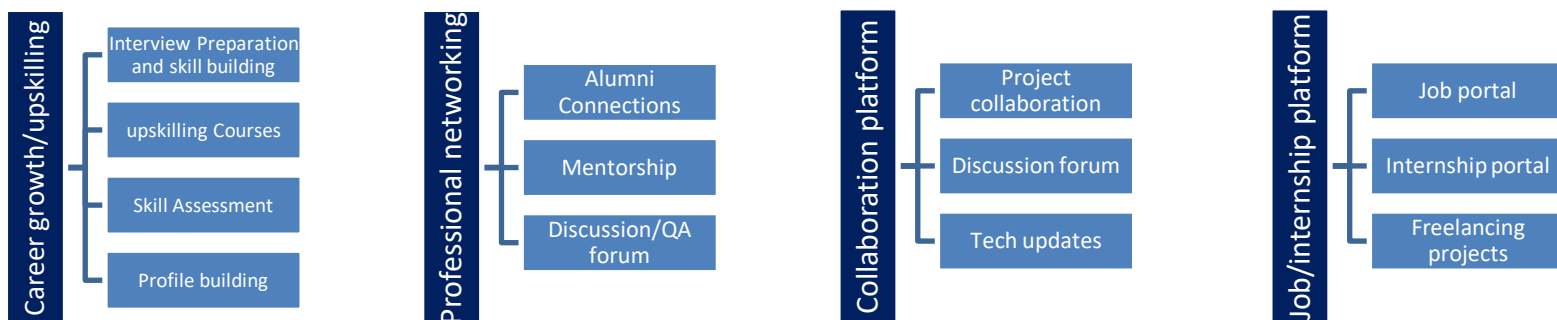


2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.





2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Scikit-learn Documentation. <https://scikit-learn.org/>
- [2] Pandas Documentation. <https://pandas.pydata.org/>
- [3] NumPy Documentation. <https://numpy.org/doc/>

2.6 Glossary

Terms	Acronym
Machine Learning	ML
Data Analysis	DA
Exploratory Data Analysis	EDA
R-Squared Score	R ²
DataFrame	DF

Problem Statement

Agriculture plays a vital role in the economy, and accurately predicting crop yield is essential for farmers, policymakers, and agricultural organizations. Crop yield depends on several factors such as crop type, cultivation cost, production cost, geographical location, and farming practices. Traditional methods of estimating crop yield are often manual, time-consuming, and less accurate.

The objective of this project is to develop a **Machine Learning-based Crop Yield Prediction System** that can accurately predict agricultural crop yield using historical agricultural data. The system uses features such as **crop type, state, cost of cultivation (A2+FL), cost of cultivation (C2), and cost of production** to estimate the expected crop yield per hectare.

The project involves collecting and preprocessing the dataset, performing exploratory data analysis (EDA), encoding categorical features, training a **Random Forest Regressor** model, and evaluating its performance using **R² Score, Mean Absolute Error (MAE), and Mean Squared Error (MSE)**. The trained model can also predict crop yield for new user inputs, helping farmers and agricultural planners make informed decisions.

The internship program was well organized by UCT and upskill Campus. During the initial phase, we understood the project requirements and problem statement. We then collected and analyzed the dataset, performed data preprocessing, built and evaluated the machine learning model, improved the prediction performance, and finally completed the project documentation and report submission.

I express my sincere gratitude to **upskill Campus, The IoT Academy, UniConverge Technologies Pvt. Ltd. (UCT)**, the faculty members of **Sathyabama Institute of Science and Technology**, and everyone who guided and supported me throughout this internship. Their continuous encouragement and valuable guidance helped me successfully complete this project.

Throughout this internship, I improved my problem-solving ability, programming knowledge, analytical thinking, and understanding of the complete Machine Learning project lifecycle. I also learned the importance of teamwork, time management, documentation, and continuous learning while working on an industry-oriented project.

The developed system achieved an **R² Score of 0.9463**, indicating that the model predicts crop yield with high accuracy on the given dataset.

3 Existing and Proposed solution

Provide summary of existing solutions provided by others, what are their limitations?

What is your proposed solution?

What value addition are you planning?

3.1 Code submission (Github link)

<https://github.com/SHAIKSUBHANI-28/upskillcampus.git>

https://github.com/SHAIKSUBHANI-28/upskillcampus/blob/main/Crop_Yield_Prediction.ipynb

3.2 Report submission (Github link) :

[https://github.com/SHAIKSUBHANI-28/upskillcampus/blob/main/Industrial Internship Report SUBHANI SHAIK USC UCT.pdf](https://github.com/SHAIKSUBHANI-28/upskillcampus/blob/main/Industrial_Internship_Report_SUBHANI_SHAIK_USC_UCT.pdf)

4 Proposed Design/ Model

4.1 The proposed system is a Machine Learning–based Agriculture Crop Production Prediction model that predicts the crop yield based on agricultural cost parameters and location. The system takes crop type, state, cultivation cost, and production cost as input. The data is preprocessed, categorical values are encoded using Label Encoder, and a Random Forest Regressor model is trained using historical agricultural data. The trained model predicts the expected crop yield in Quintal per Hectare, helping farmers and agricultural planners make informed decisions.

4.2 High Level Diagram (if applicable)

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

4.3 Low Level Diagram (if applicable)

4.4 Interfaces (if applicable)

Update with Block Diagrams, Data flow, protocols, FLOW Charts, State Machines, Memory Buffer Management.

5 Performance Test

The performance of the Agriculture Crop Production Prediction system was evaluated using a historical agricultural dataset. The model was trained using the Random Forest Regression algorithm and tested on unseen data to evaluate its prediction accuracy. Important constraints such as prediction accuracy, processing time, memory usage, and model reliability were considered during testing. The Random Forest model provided accurate predictions with minimal processing time, making it suitable for agricultural prediction tasks

5.1 Test Plan/ Test Cases

Test Case	Input	Expected Result	Actual Result	Status
TC-01	Valid crop and state details	Predict crop yield	Crop yield predicted successfully	Pass
TC-02	Different cultivation cost values	Updated prediction	Prediction changed correctly	Pass
TC-03	Different production cost	Correct yield prediction	Prediction generated successfully	Pass
TC-04	Multiple crop types	Accurate prediction	Model predicted successfully	Pass
TC-05	Large input values	Stable prediction	No system failure	Pass

5.2 Test Procedure

1. Load the agricultural dataset into the Python environment.
2. Perform data preprocessing and label encoding.
3. Split the dataset into training and testing datasets.
4. Train the Random Forest Regressor model.
5. Test the model using unseen test data.
6. Calculate model performance metrics.
7. Compare predicted values with actual crop yield values.
8. Save the trained model using the Pickle library.

5.3 Performance Outcome

The Random Forest Regression model achieved good prediction performance on the agricultural dataset.

Metric	Value
Algorithm	Random Forest Regressor
Training Samples	39
Testing Samples	10
R ² Score	0.9463
Mean Absolute Error (MAE)	28.44
Mean Squared Error (MSE)	4852.74

6 My Learnings

During this internship, I gained practical knowledge of Machine Learning by working on the **Agriculture Crop Production Prediction** project. I learned how to collect and preprocess datasets, perform data analysis using Pandas and NumPy, and visualize data using Matplotlib.

I understood the complete Machine Learning workflow, including data cleaning, feature selection, label encoding, train-test splitting, model training, evaluation, and prediction. I implemented the **Random Forest Regression** algorithm to predict crop yield and evaluated the model using performance metrics such as **R² Score, Mean Absolute Error (MAE), and Mean Squared Error (MSE)**.

I also learned how to save trained models using the **Pickle** library, create a Jupyter Notebook project, prepare technical documentation, and upload the project to GitHub. This internship improved my problem-solving skills, programming knowledge, and understanding of real-world data analysis.

Overall, this internship provided valuable hands-on experience in Data Science and Machine Learning, which will help me build more advanced predictive models and contribute to my future career as a Machine Learning Engineer and Data Scientist.

In addition, I learned how to save trained Machine Learning models using the **Pickle** library, make predictions using new input data, and organize project files effectively. I also gained experience in documenting the project, preparing a technical report, and publishing the project on **GitHub**, which improved my software development and version control skills.

I gained hands-on experience in collecting and understanding datasets, performing data preprocessing, handling missing values, and preparing data for model training. I learned to use Python libraries such as **Pandas, NumPy, Matplotlib, and Scikit-learn** for data manipulation, visualization, and model development.

During the project, I understood the importance of feature selection, label encoding, splitting data into training and testing sets, and training a **Random Forest Regressor** model. I also learned how to evaluate model performance using metrics such as **R² Score, Mean Absolute Error (MAE), and Mean Squared Error (MSE)** to measure prediction accuracy.

This internship enhanced my analytical thinking, problem-solving ability, programming skills, and understanding of the complete Machine Learning workflow. The knowledge and experience gained during this internship have strengthened my confidence and will help me in future academic projects and in building a successful career in **Machine Learning, Data Science, and Artificial Intelligence**.

7 Future Work Scopes

Although the developed Agriculture Crop Production Prediction system provides accurate crop yield predictions, there are several opportunities for future improvements. Due to time and resource limitations, some advanced features could not be implemented during this internship.

Future enhancements may include:

- Integrating real-time weather data such as rainfall, temperature, and humidity to improve prediction accuracy.
- Including soil characteristics such as pH value, moisture content, and nutrient levels for better agricultural analysis.
- Expanding the dataset to cover more crops, states, districts, and seasonal variations across India.
- Developing a web or mobile application to provide farmers with easy access to crop yield predictions.
- Comparing multiple Machine Learning algorithms such as XGBoost, Gradient Boosting, and Artificial Neural Networks to identify the best-performing model.
- Implementing deep learning techniques for handling large-scale agricultural datasets.
- Providing fertilizer and irrigation recommendations along with crop yield prediction.
- Integrating Geographic Information System (GIS) and satellite imagery to enhance prediction accuracy.
- Supporting multiple regional languages to make the application more user-friendly for farmers.
- Deploying the model on cloud platforms to enable real-time predictions and remote accessibility.

These enhancements can improve the system's accuracy, usability, and scalability, making it more beneficial for farmers, agricultural researchers, and government agencies in planning and decision-making.



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